

Gate 5A - Distributed Data

Protected first attempt | Parcel telemetry | RC1

Scenario

A parcel network receives historical scan files plus a continuous event stream. Build a scalable PySpark/Delta design that produces trusted latest-shipment state and hub metrics, supports an incremental update safely, and explains Kafka partition/offset/replay behavior.

Required work

- Inspect schema/grain before coding.
- Use explicit PySpark schema.
- Clean invalid records and deduplicate business events.
- Join hub reference data and compute latest shipment state with a window.
- Write appropriate Parquet/Delta output with a justified partition strategy.
- Apply incremental updates with merge/upsert semantics; rerun must not duplicate logical records.
- Inspect/explain one shuffle/performance risk.
- Complete Kafka/Structured Streaming starter with topic/key/group/checkpoint reasoning.
- Run offline checker and save evidence.

Protected rules

- Documentation allowed.
- No direct AI solution until attempt is saved.
- Do not open Gate 5A Review before saving the attempt.
- Live Spark/Delta/Kafka runtime evidence is required later for FINAL.